LIMIT PRESERVING REPRESENTABLE FUNCTORS ARE LOCALLY OF FINITE PRESENTATION

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 4857-4862 of sources/stacks-artin/pinned-original-source.tex; proof span is lines 4864-4880, the complete author proof (15 body lines).

Editorial note. Navigation stubs: 2 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Queue index 38 of queue_artin.json (pb=951, stmt 4857-4862, proof 4864-4880).

Editorial note. stacks-artin wave ar01 (2026-10-14): fresh adjudication accept per lane-s3/artin_judgments.md; pool per Chief 10:5x direct dispatch (idle-first over S12's 01:5x note, SMC precedent); artin.tex 99 rows/67 with proof judged 46 accept / 21 reject / 5 AGP-overlap avoidance rows; patch-sd-envs contract spans scanned.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. The cited inputs are prerequisites; the argument in the retained span is not replaced by any cited result.

Source reference. Original lemma 00QN: sources/stacks-artin/pinned-original-source.tex, source lines 4880–4880. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 01Z4: sources/stacks-artin/pinned-original-source.tex, source lines 4880–4880. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let X be an object of $(\mathit{Noetherian}/S)_\tau$. If the functor of points $h_X : (\mathit{Noetherian}/S)_\tau^{opp} \to \mathit{Sets}$ is limit preserving, then X is locally of finite presentation over S.*

Proof. Let $V \subset X$ be an affine open subscheme which maps into an affine open $U \subset S$. We may write $V = \lim V_i$ as a directed limit of affine schemes V_i of finite presentation over U, see Algebra, Lemma 00QN. By assumption, the arrow $V \to X$ factors as $V \to V_i \to X$ for some i. After increasing i we may assume $V_i \to X$ factors through V as the inverse image of $V \subset X$ in V_i eventually becomes equal to V_i by Limits, Lemma 01Z4. Then the identity morphism $V \to V$ factors through V_i for some i in the category of schemes over U. Thus $V \to U$ is of finite presentation;

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the corresponding algebra fact is that if B is an A-algebra such that $\mathrm{id} : B \to B$ factors through a finitely presented A-algebra, then B is of finite presentation over A (nice exercise). Hence X is locally of finite presentation over S. $\square$